

Tuning filesystem

Filesystem info

`dumpe2fs [options] device`
 info on ext2, ext3, filesystems
device is the filesystem device file

example

`sudo fdisk -l`

```
Device            Start      End          Sectors      Size Type
/dev/nvme0n1p1      34        32767        32734       16M Microsoft reserved
/dev/nvme0n1p2    32768    1083391    1050624     513M EFI System
/dev/nvme0n1p3   1083392   977303551   976220160   465.5G Microsoft basic data
/dev/nvme0n1p4   977303552 1953521663   976218112   465.5G Linux filesystem
```

device file: `/dev/nvme0n1p4`

```
james@laptop:~$ sudo dumpe2fs -h /dev/nvme0n1p4
dumpe2fs 1.47.0 (5-Feb-2023)
Filesystem volume name:   <none>
Last mounted on:         /
Filesystem UUID:         79455c4b-f6ce-47d7-9557-227040be4112
Filesystem magic number: 0xEF53
Filesystem revision #:    1 (dynamic)
Filesystem features:      has_journal ext_attr resize_inode dir_index
Filesystem flags:         signed_directory_hash
Default mount options:    user_xattr acl
Filesystem state:         clean
Errors behavior:          Continue
Filesystem OS type:       Linux
```

Tuning filesystem parameters

`tune2fs [options] device`

options

<code>-c mounts</code>	adjust the maximum mount count a disk can be mounted up to the maximum mount count without being checked with <code>fsck</code>
<code>-i interval</code>	adjust the time between disk checks
<code>-j</code>	add a journal to a filesystem converts a ext2 into an ext3 filesystem
<code>-m percent</code>	set reserved blocks a percentage of disk space reserved for use by <i>root</i>

Mount Point

- A directory in the filesystem where you attach or disconnect another filesystem
- When the filesystem is mounted, its contents become available
- e.g. plugging in a USB drive will create a directory `/mnt/usb_photos`` which is your mount point

Debugging filesystem

debugfs

view and modify a filesystem

similar to *dumpe2fs*, *tune2fs*

prompts for commands:

<code>show_super_stats stats</code>	show superblock information
<code>stat [filename]</code>	show inode data on a file
<code>undelete inode name</code>	un-delete a file

superblock

- like a table of contents for a filesystem
- contains information the OS needs to sue it

inode

- index node
- stores metadata about a file or directory
- doesn't include it's name and content
- e.g. file type, permissions, size etc

Journal

What is a Journal

- a file that records changes it intends to make **before** committing those changes to the main file system
- it's like a "to-do" list

Prevents inconsistencies

1. if there's a crash/power failure, the system can check and journal and examine data structures described in it
 2. if there are inconsistencies, it can do a roll-back
- Journaling filesystems are standard in most partitions

Checking for a journal

```
james@laptop:~$ sudo dumpe2fs /dev/nvme0n1p4 | grep journal
dumpe2fs 1.47.0 (5-Feb-2023)
Filesystem features:      has_journal ext_attr resize_inode dir_index filetype needs_recovery extent 64bit flex_bg sparse_super large_file huge_file dir_nlink extra_isize metadata_csum
Journal features:        journal_incompat_revoke journal_64bit journal_checksum_v3
Total journal size:      1024M
Total journal blocks:    262144
```

tune2fs

to modify the journal:

<code>-J size=journal-size</code>	set journals size
<code>-J device=external-journal</code>	set device on which it's stored

Checking for errors

Bugs, power failures and mechanical problems can corrupt data structures on a filesystem

fsck

uses tools like `e2fsck`

only use on unmounted filesystems

`fsck [options] filesystem`

options:

<code>-A</code>	check all files
<code>-C</code>	show progress
<code>-N</code>	no action just describe what it would normally do

Disk Space

`df` check disk use by partition

`du` space by directory

`df [options] [files]`

<code>a --all</code>	all filesystems
<code>-h --human-readable</code>	use scaled units e.g. instead of 5859784 blocks shows 5.6G (for 5.6GiB)
<code>-l --local</code>	local filesystems only not network filesystems
<code>-T --print-type</code>	shows filesystem type
<code>-t [fstype] --type=fstype</code>	only show this filesystem type

example

```
james@laptop:~$ df -hl /dev/nvme0n1p4
Filesystem      Size  Used Avail Use% Mounted on
/dev/nvme0n1p4 458G 144G 291G  34% /
```

`du [options] [directories]`

how much disk space a directory is using

is recursive

options:

<code>-a --all</code>	reports of individual file sizes as well
<code>-c --total</code>	show total
<code>-h --human-readable</code>	same as <code>df</code>
<code>-s --summarize</code>	summarize subdirectories
<code>-x --one-file-system</code>	limit to current filesystem

```
james@laptop:~$ du -chsx ~/Downloads
45M    /home/james/Downloads
45M    total
```